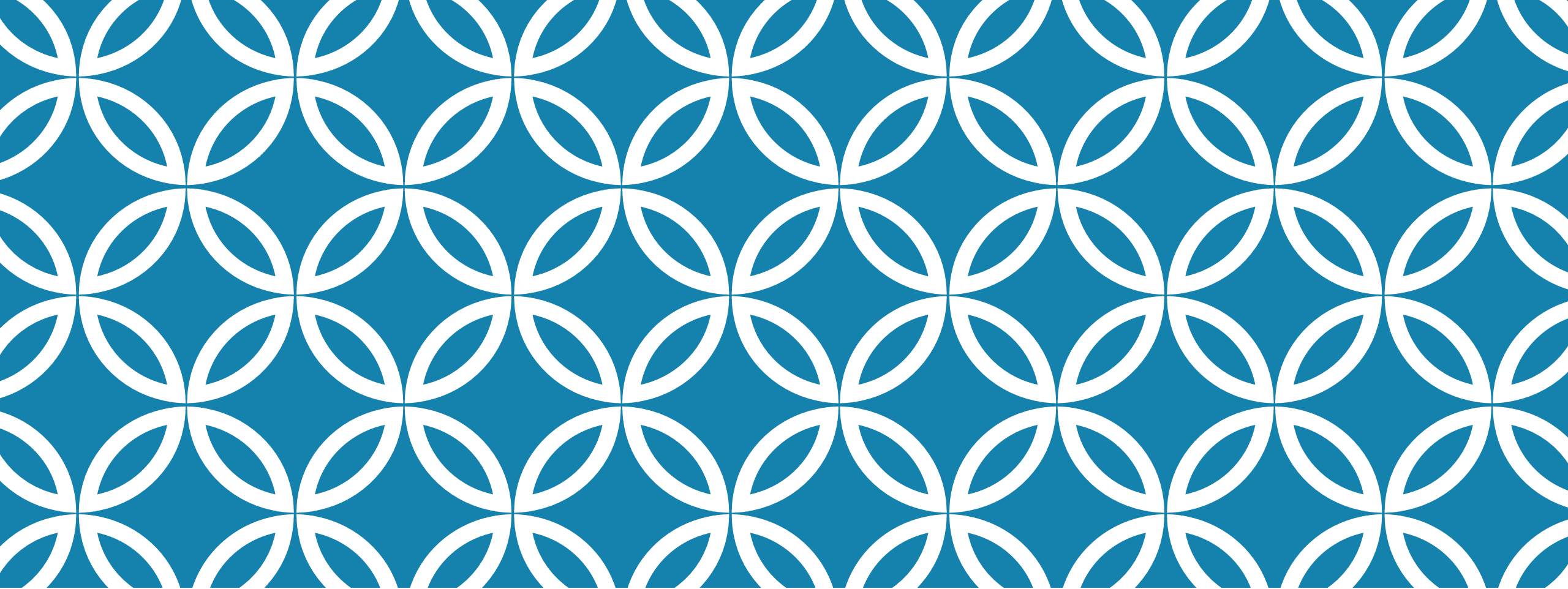


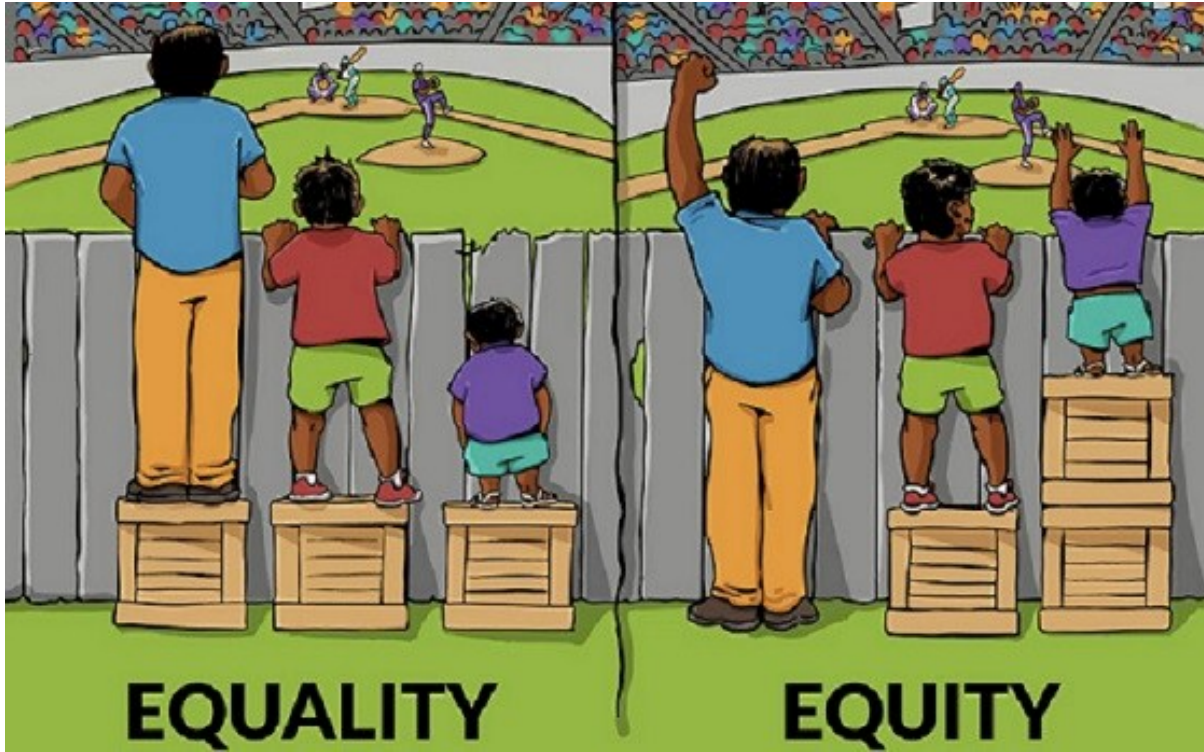


PUBP309: EDUCATION POLICY

Educational Equity: Theory and Measurement

Shivakumar Jolad
FLAME University

EQUITY AND EQUALITY



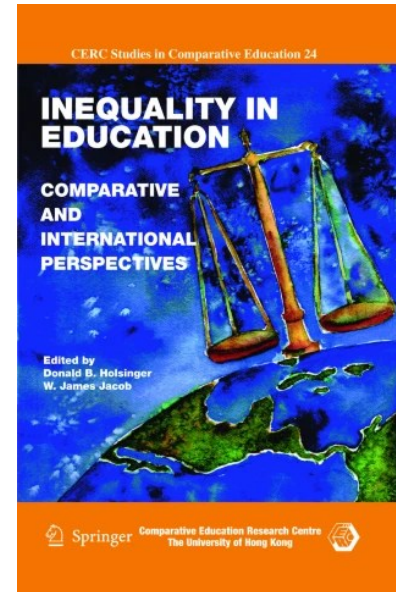
Equality as “the state of being equal in terms of quantity, rank, status, value or degree”,

Equity “considers the social justice ramifications of education in relation to the **fairness, justness and impartiality of its distribution** at all levels or educational sub-sectors”.

Accident of birth should not affect the education opportunity, process or outcome

<http://www.publichealthnotes.com/equity-vs-equality/>

Jacob, W. J. and D. B. Holsinger (2008). “Inequality in education: a critical analysis”. In D. B. Holsinger and W. J. Jacob (eds.), *Inequality in Education: Comparative and International Perspectives*,



CASTE, GENDER AND LITERACY IN 1901

AN UNCERTAIN GLORY

Table 8.1
Caste, Gender and Literacy in 1901

Region (Province or State)	Literacy Rates, 1901 (%) Brahmin ^a		Literacy Rates, 1901 (%) 'Scheduled Castes' ^b	
	Male	Female	Male	Female
Baroda State	73.0	5.6	1.2	0
Mysore State	68.1	6.4	0.9	0.1
Bombay Province	58.0	5.4	0.7	0
Madras Province	57.8	4.4	1.0	0
United Provinces	55.3	4.6	0.2	0
Central Provinces	36.5	0.9	0.4	0



BRIDGING GAP

Figure -1: Gross Enrolment Rate (Higher Education) : 2015- 16

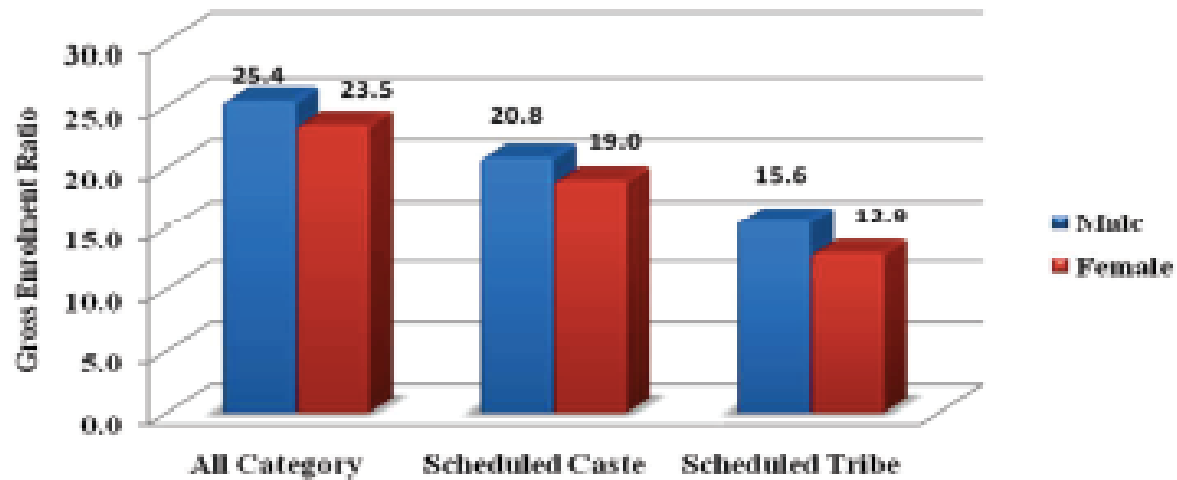
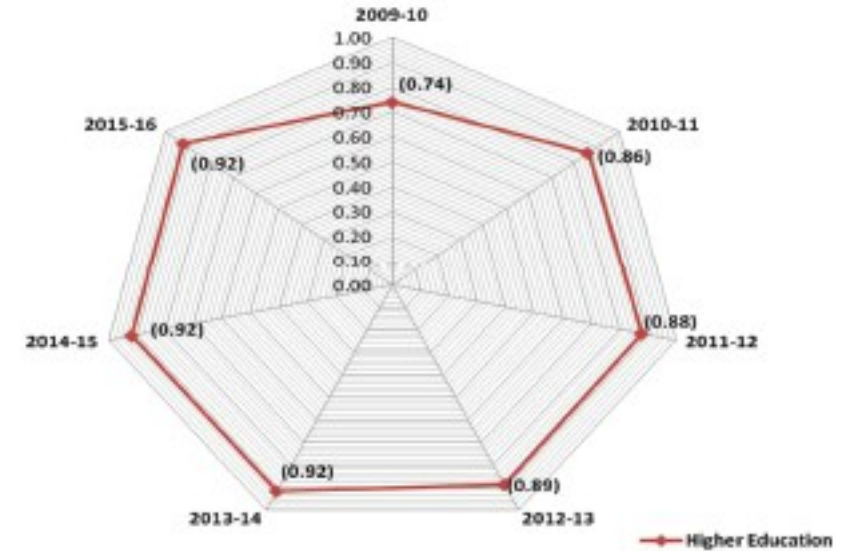


Figure -2 Gender Parity Index at Higher Education over the years (2009 to 2016)



MHRD Education statistics 2017-18

FRAMEWORK TO UNDERSTAND EQUITY

Why should we be concerned about equity?



Equity cannot be divorced from **a normative framework about fairness and justice.**



Layers: educational inputs, processes , and outcomes

UN Convention on rights of Child-Article 2: All children have these rights, no matter who they are, where they live, what their parents do, what language they speak, what their religion is, whether they are a boy or girl, what their culture is, whether they have a disability, whether they are rich or poor. No child should be treated unfairly on any basis.

FACTORS AFFECTING DIFFERENCE IN LEARNING OUTCOME

- ❑ Individual differences in **ability and motivation**,
- ❑ type of **family and social background** one comes from
- ❑ type of resources one has access – **Institutional**/**environmental** factors

“all differences in **distribution of effort between types are morally arbitrary**, while individuals can be held responsible for their degree of effort relative to other members of their type.

EQUALITY OF OPPORTUNITY

- ❑ Same opportunity irrespective of accident of birth or environment he/she grew up
- ❑ **Remove the barriers to access**
- ❑ How do you decide whether there were equality of opportunity?
- ❑ How do you distinguish EOP between groups, and between individuals?
- ❑ Does EoP imply **equality of outcome**?
 - Is **equality of outcome desirable** in society?

ARGUING FOR AFFIRMATIVE ACTION

□ Justice at Harvard, lecture 17, Arguing for Affirmative action-



Harvard University's

JUSTICE

with Michael Sandel

<https://www.youtube.com/watch?v=2ZMxveNmROw>

JUSTICE AS FAIRNESS

❑ Rawls 'A theory of Justice'

- Veil of Ignorance
- Difference principle

❑ SEE: Justice at Harvard- Lecture 14



GENDER AND EQUITY

INEQUALITY PLAYS OUT IN DIFFERENT WAYS IN EDUCATION:

Who goes to what kind of school and why

Who moves from primary to upper primary and secondary

Why some children emerge barely literate after years of schooling

Schooling experience of the very poor and most disadvantaged – especially girls – curriculum, textbooks and teacher-student interactions

Responsiveness of the system, factors that facilitate / impede

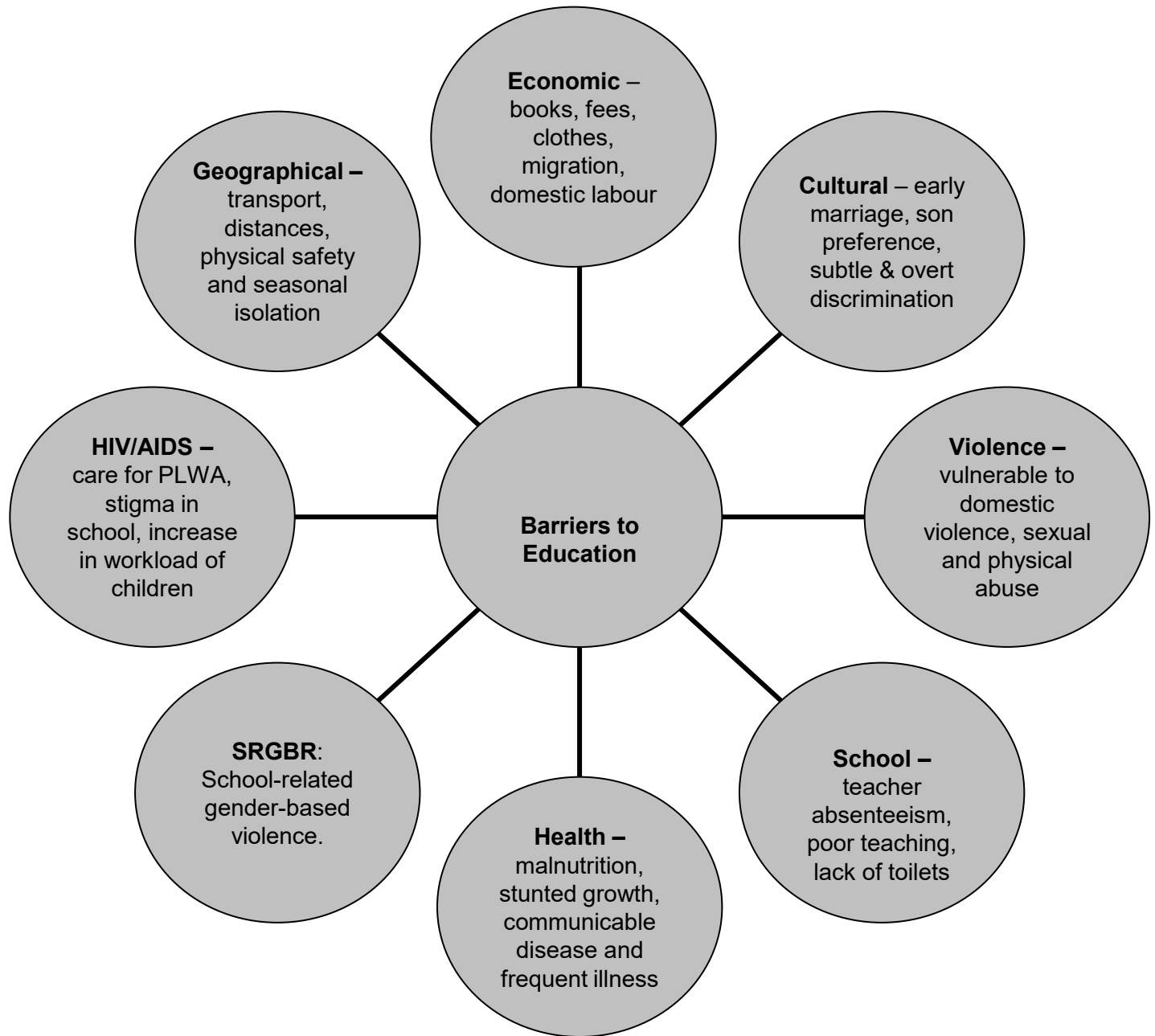
Who is included / excluded – when, why and where



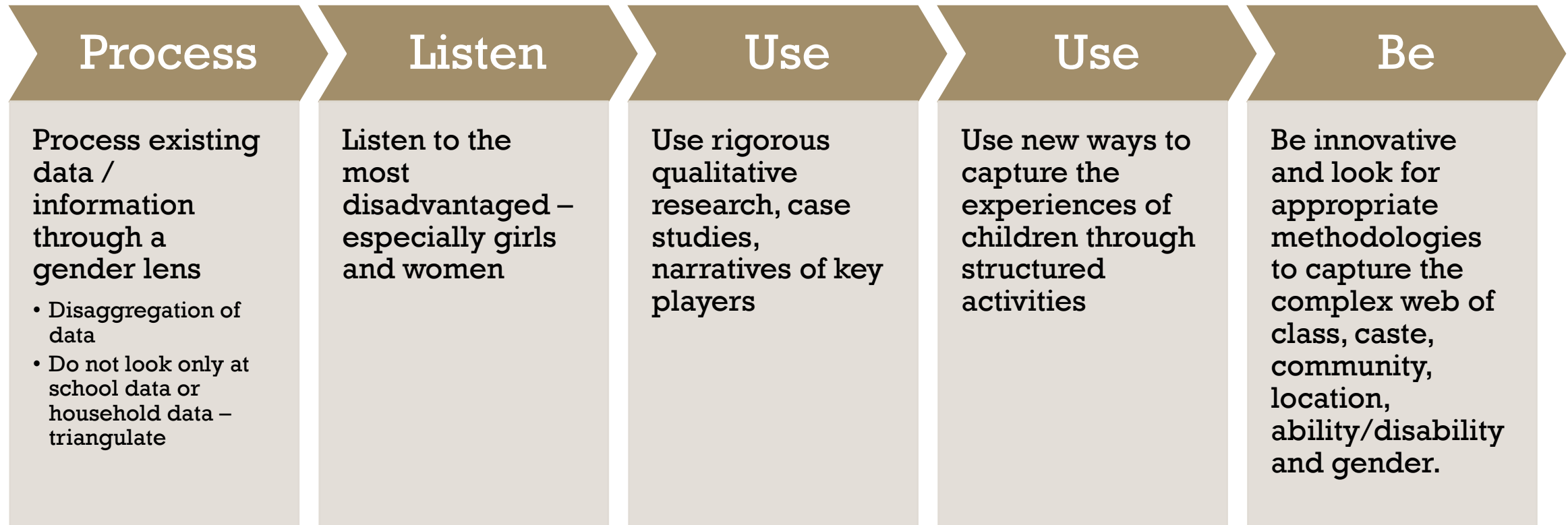
Systemic Issues	Content and Process of Education	Economy, Society and Culture
⇒ Problem of access ⇒ Dysfunctional schools ⇒ Quality of schools ⇒ Motivation of teachers ⇒ Existence of multiple delivery systems - formal, non-formal, condensed, satellite and residential ⇒ Calendar and timings	⇒ Gender stereotyping ⇒ Perpetuation of gender-bias ⇒ Relevance of curriculum ⇒ Language ⇒ Joyful learning ⇒ Ready access to books, magazines, papers and so on ⇒ Appropriate reading material for the newly literate	⇒ Poverty / powerlessness ⇒ Status of women ⇒ Cost to family ⇒ Child labour / domestic chores ⇒ Farm / non-farm work ⇒ Caught up in survival battles ⇒ Perception of herself ⇒ Post puberty practices ⇒ Child marriage ⇒ Conflict, violence in community / family



BARRIERS TO EDUCATION



WHAT AND HOW





MEASURING EQUITY



BASIC CLASSIFICATION

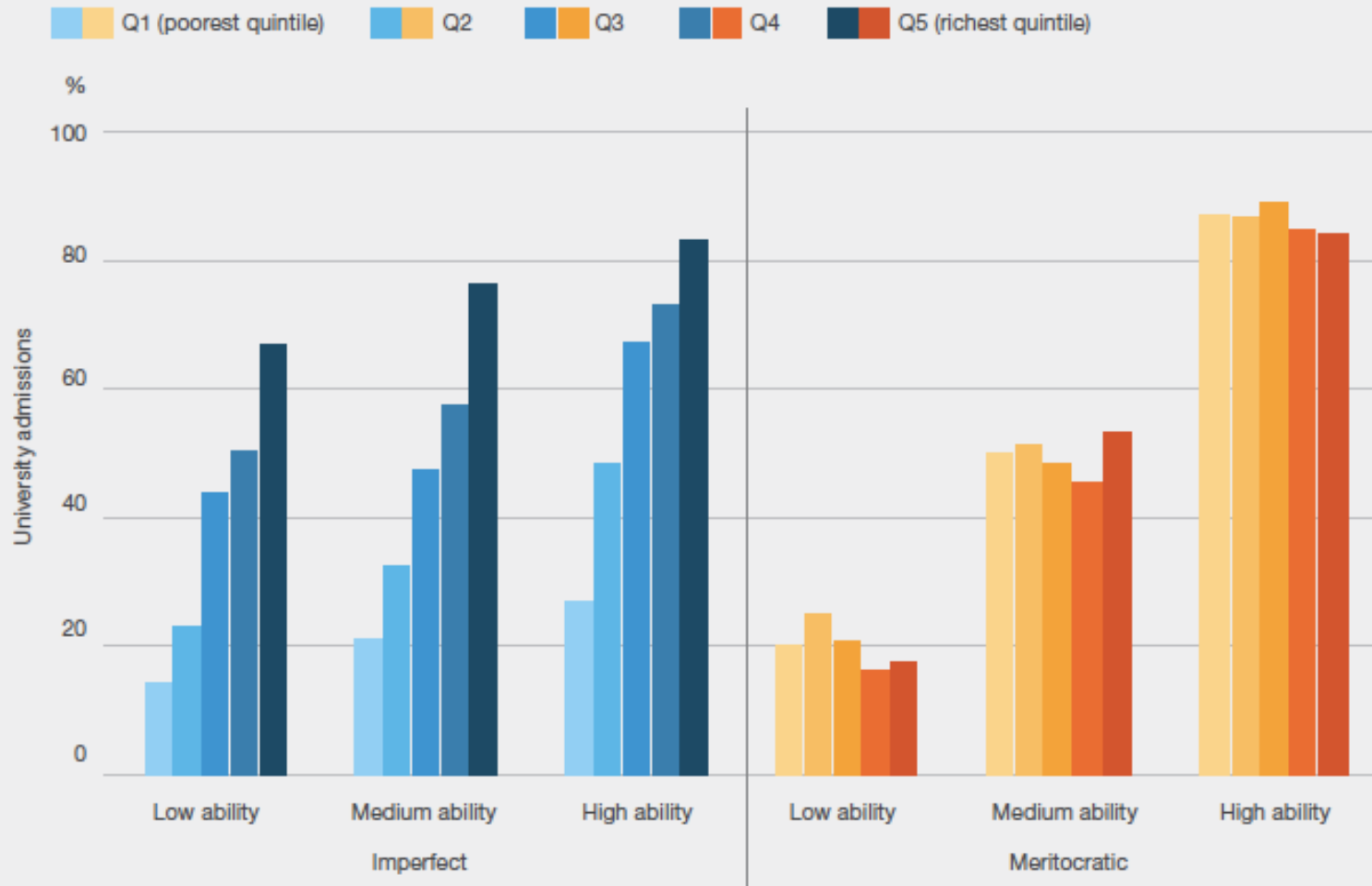
- ❑ **Univariate:** based on the distribution of an educational variable
- ❑ **Bivariate/multivariate:** based on the joint distribution of an educational variable and one or more characteristics

Table 2.1 Classification of equity concepts and related equity norms

Univariate based on the distribution of an educational variable	Bivariate/multivariate based on the joint distribution of an educational variable and one or more characteristics
Minimum standards binary educational variable (e.g. completed primary education) is positive for everyone	Impartiality education does not depend on background characteristics
Equality of condition educational variable is the same for everyone	Meritocracy education is positively related to ability but not related to other characteristics
	Redistribution education is positively related to disadvantage

MERITOCRACY

Figure 2.1 Imperfect and perfectly meritocratic distribution of university admissions



Note: Hypothetical data.

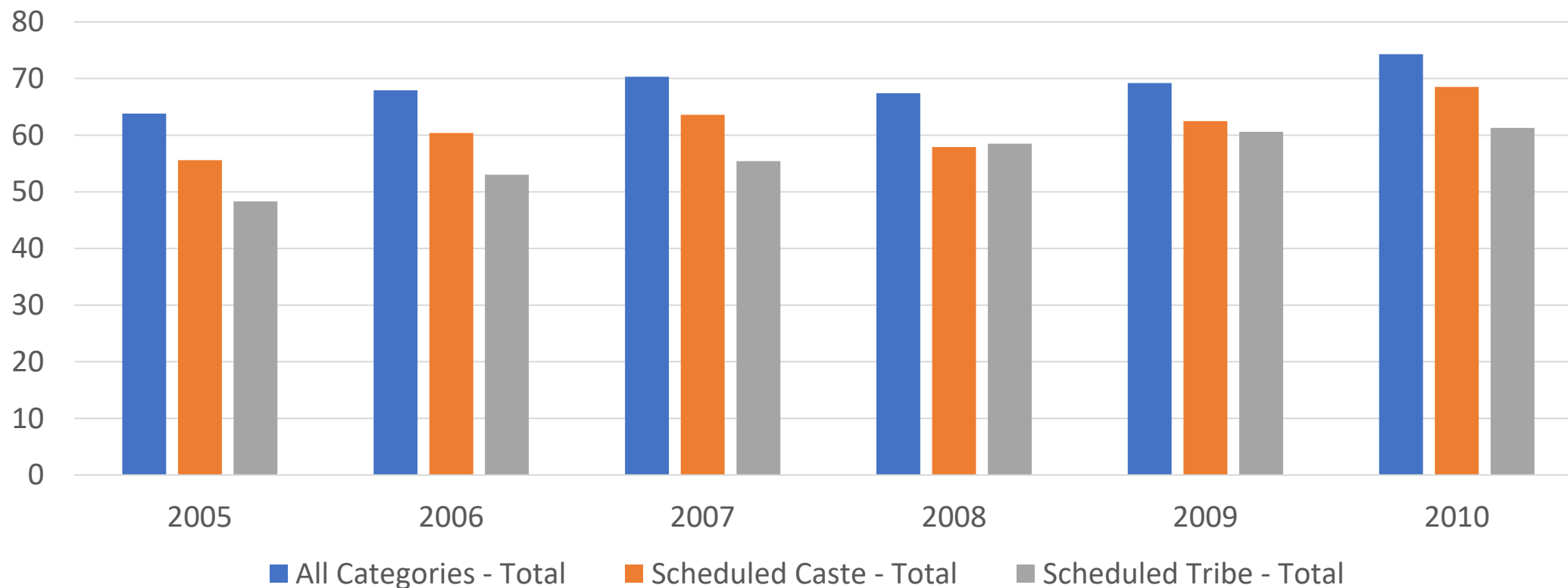
educational opportunities are distributed on the basis of merit

Admissions depend on income or ability?

HIGH SCHOOL PASS PERCENTAGE

High school examination pass percentage during 2005-2010

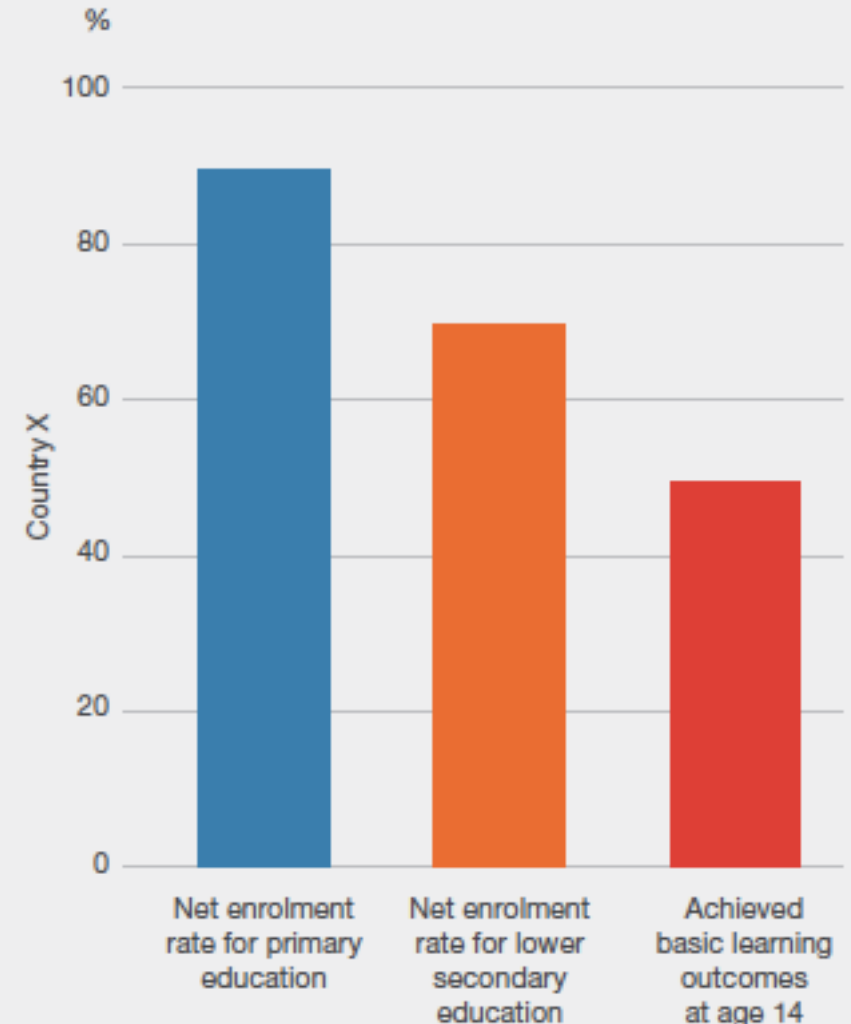
Source : MHRD- education statistics



MINIMUM STANDARDS

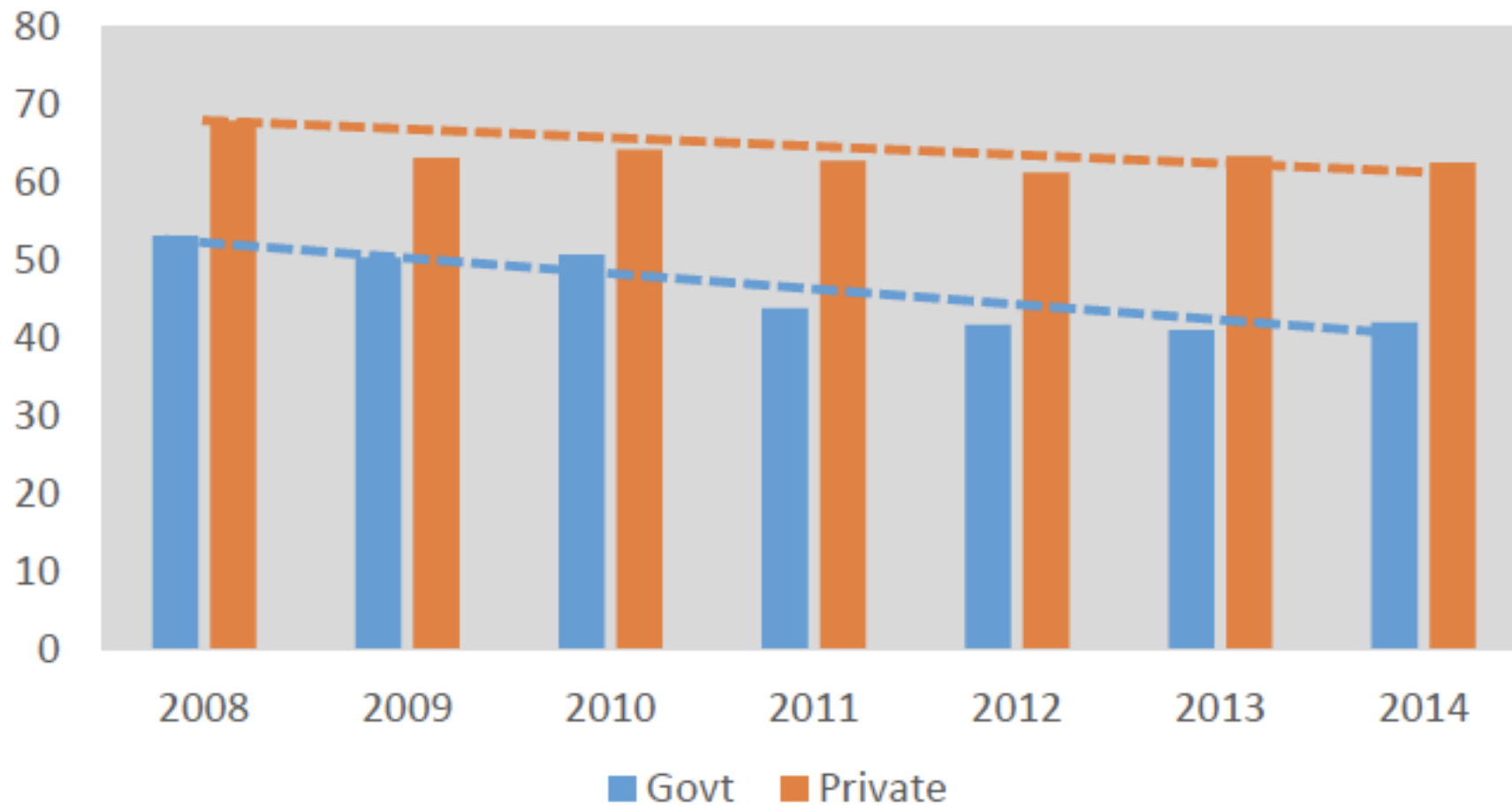
- ❑ Education in terms of a binary criterion
 - a child is enrolled in primary school or not,
 - can demonstrate basic literacy
- ❑ measuring the proportion of individuals who meet the minimum standard could be taken as an equity measure
- ❑ **equity is achieved when 100% of individuals meet the standard.**
- ❑ Millennium Development goals-2015

Figure 2.2 Proportion of children reaching three types of minimum standards



Note: Hypothetical data.

% Children in Std V who can read Std II level text
ASER 2008-2014
All India (rural)



Annual Status of Education Report
असर
ASER **2014** **RURAL**
Facilitated by PRATHAM

IMAPRTIALITY

- ❑ It argues that educational goods should be **distributed equally** with respect to differences which should be irrelevant, **such as gender, race, wealth or location**
 - Hold accountable on only those within their control
- ❑ **absence of any relationship between an education indicator of interest and one or more measures of circumstance**

GAP DIFFERENCE

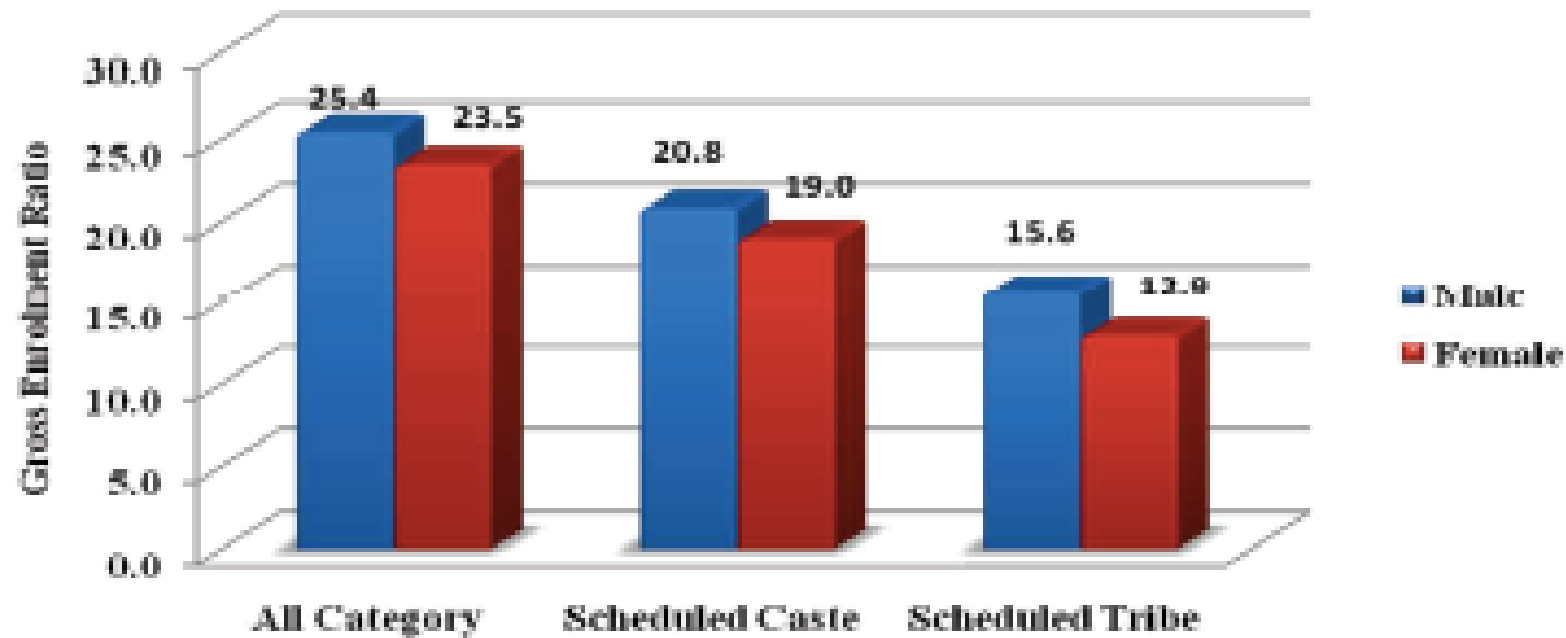
Family of metrics

Gap, difference

Impartiality measures
and analyses

Cross-tabulation or
disaggregated bar
charts

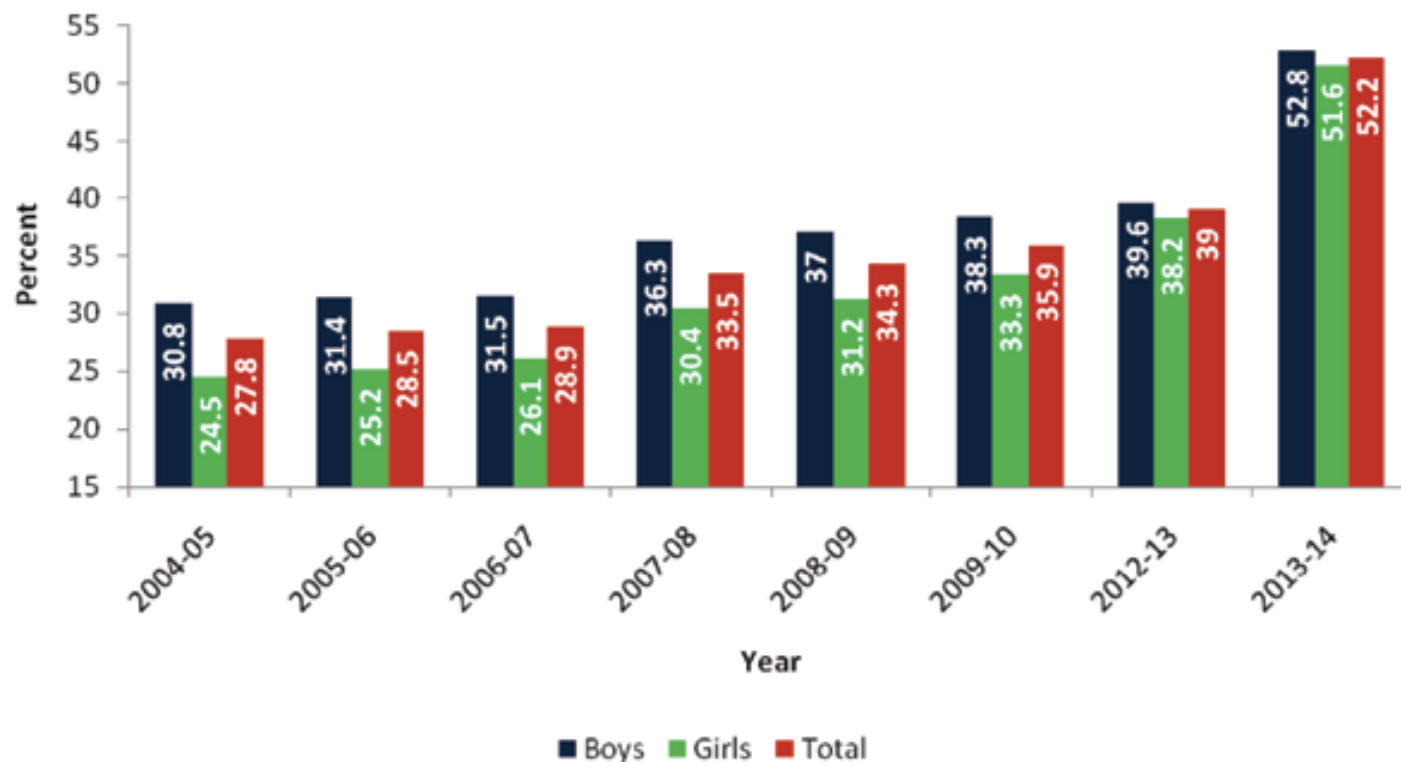
Figure -1: Gross Enrolment Rate (Higher Education) : 2015 - 16



MHRD Education statistics

GER IN SECONDARY EDUCATION

Figure 2.3.2: Gross Enrolment Ratio (GER) in higher secondary education
(Classes XI-XII) (Age 16-17 Years) (All categories of students)



Source: Educational Statistics at a Glance, 2011, MHRD, GoI; Statistics in School Education, 2010-11, MHRD, GoI; and U-DISE, NUEPA.

Education For All
Towards Quality with Equity
INDIA

Table 3: Literacy rates by state, area, and gender, census data

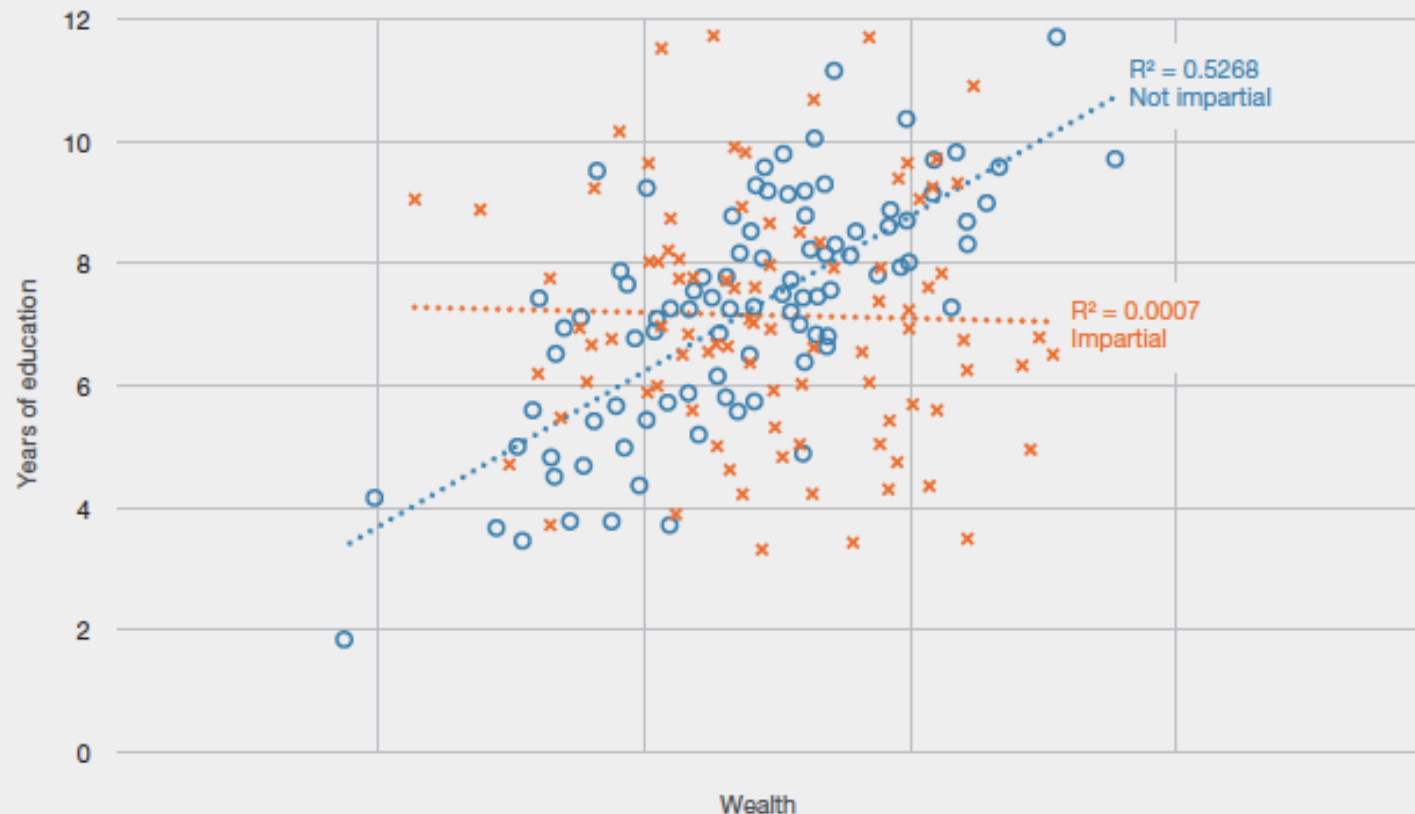
	Male			Female			Persons		
	1991	2001	Increase	1991	2001	Increase	1991	2001	Increase
Andhra Pradesh	55.1	70.9	15.8	32.7	51.2	18.5	44.1	61.1	17.0
Bihar	52.5	62.2	9.7	22.9	35.2	12.3	38.5	49.2	10.7
Gujarat	73.1	76.5	3.4	48.6	55.6	7.0	61.3	66.4	5.1
Haryana	69.1	79.3	10.2	40.5	56.3	15.8	55.9	68.6	12.7
Himachal Pradesh	75.4	84.6	9.2	52.1	67.1	15.0	63.9	75.9	12.0
Karnataka	67.3	76.3	9.0	44.3	57.5	13.2	56.0	67.0	11.0
Kerala	93.6	94.2	0.6	86.1	87.9	1.8	89.8	90.9	1.1
Madhya Pradesh	58.4	77.0	18.6	28.9	51.0	22.1	44.2	64.4	20.2
Maharashtra	76.6	86.3	9.7	52.3	67.5	15.2	64.9	77.3	12.4
Orissa	63.1	76.0	12.9	34.7	51.0	16.3	49.1	63.6	14.5
Punjab	65.7	75.6	9.9	50.4	63.6	13.2	58.5	70.0	11.5
Rajasthan	55.0	76.5	21.5	20.4	44.3	23.9	38.6	61.0	22.4
Tamil Nadu	73.8	82.3	8.5	51.3	64.6	13.3	62.7	73.5	10.8
Uttar Pradesh	55.7	70.9	15.2	25.3	43.9	18.6	41.6	58.1	16.5
West Bengal	67.8	77.6	9.8	46.6	60.2	13.6	57.7	69.2	11.5
India	64.1	75.6	11.5	39.3	54.0	14.7	52.2	65.2	13.0

	Year	GER in secondary education (Classes IX-X) (Age 14-15 years) (%)		
		Boys	Girls	Total
All	2013-14*	76.8	76.5	76.6
SC	2013-14	78.2	79.4	78.7
ST	2013-14	70.3	70.1	70.2

Source: Educational Statistics at a Glance, 2011, MHRD, GoI; Statistics in School Education, 2010-11, MHRD, GoI; and U-DISE, NUEPA.

CO-VARIATION: CORRELATION

Figure 2.4 Impartiality of education with respect to wealth



Note: Hypothetical data.

Co-variation

Correlation coefficient (r)

$$y = X\beta + \epsilon,$$

$$SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2,$$

$$SS_{\text{reg}} = \sum_i (f_i - \bar{y})^2,$$

$$R^2 \equiv 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

Source: Image from the UNESCO for
Equations from wiki

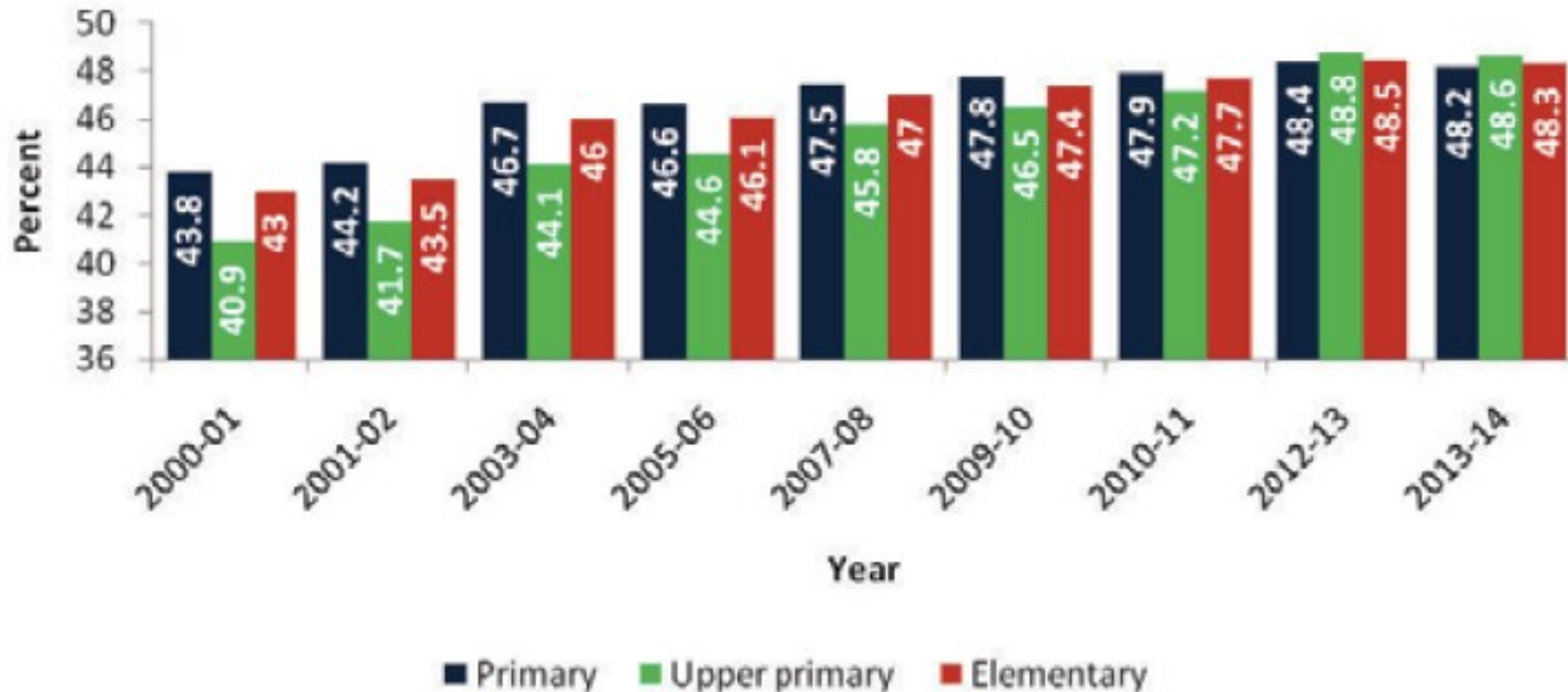
RATIOS

Ratio

Parity indices

The gender parity index is the ratio of female to male values of a given indicator. Similar parity indices can be calculated for wealth quintiles, rural and urban location, etc.

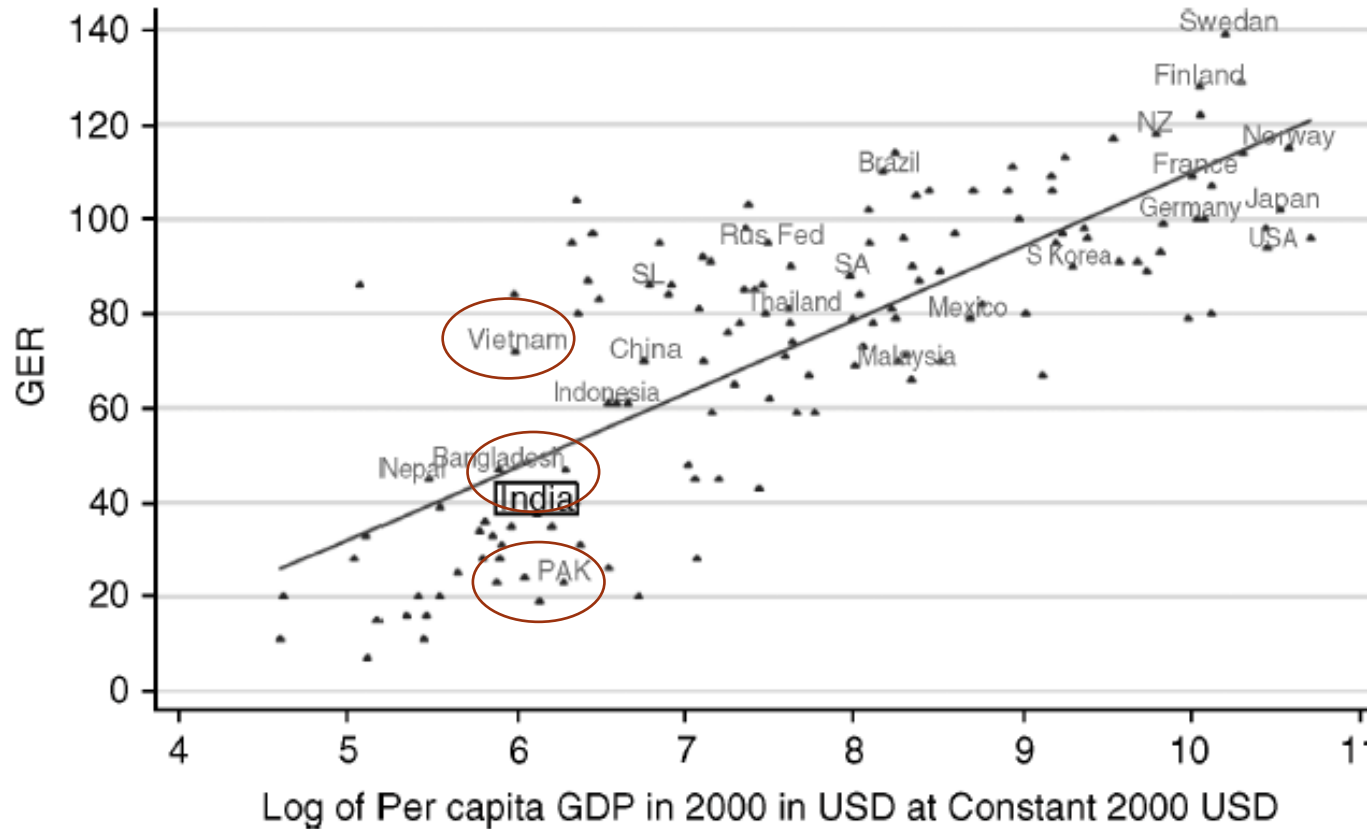
Figure 2.5.1: Girls enrolled as percentage of total enrolment in primary, upper primary and elementary education (2000-01 to 2013-14) (%)



Education For All
Towards Quality with Equity
INDIA

GER IN SECONDARY EDUCATION

Figure 1: Cross-country comparison of gross enrolment ratios in secondary education and per-capita GDP, 2000



Geeta Gandhi Kingdon
Oxford Review of Economic Policy,
Volume 23, Number 2, 2007, pp.168–195

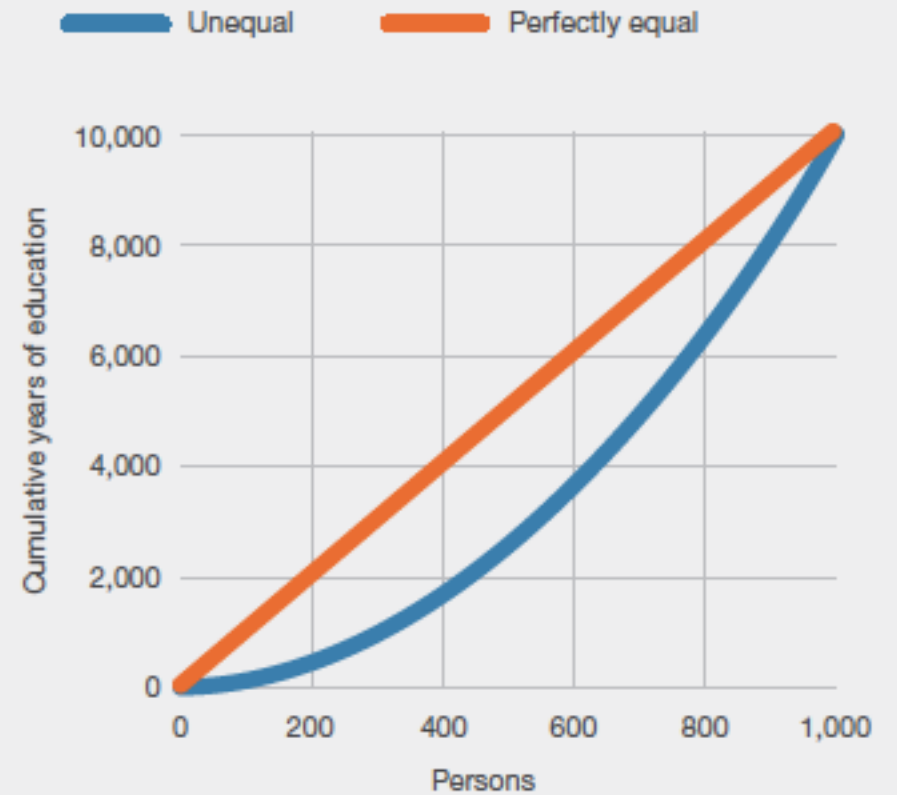
Source: World Bank (2006); calculation is based on MHRD *Selected Education Statistics* for India and World Bank's Education Statistics Database for other countries.

OTHER MEASURES OF COVARIANCE

Slope Index of Inequality (SII)	Coefficient in an ordinary least squares (OLS) or weighted least squares (WLS) regression, e.g. of years of education on wealth (Wagstaff et al., 1991). Individual or grouped (e.g. district or wealth quintile) data may be used.
Relative Index of Inequality (RII)	Slope index of inequality divided by the mean level of the dependent (education) variable (Wagstaff et al., 1991).
Elasticity	The percentage effect of a 1 % change in the characteristic (e.g. poverty) on the indicator (e.g. government education expenditure). Elasticity can be calculated from regression coefficients (see Berne and Stiefel, 1984, Table 2.4).
Proportion of variance explained by circumstances (R^2)	Uses OLS regression to examine the proportion of the variance in an indicator explained by circumstances like wealth and location. Variants use different regression specifications, or a measure of total inequality other than the variance.

EQUALITY CONDITION

Figure 2.5 Lorenz curves showing a perfectly equal, and somewhat unequal, distribution of years of education



Note: Hypothetical data.

